Spring 2002

Problem 1 (Sp02). Let U, V, W be finite dimensional subspaces of a vector space. Prove that $\dim(U)+\dim(V)+\dim(W)-\dim(U+V+W) \geqslant \max\{\dim(U\cap V),\ \dim(U\cap W),\ \dim(V\cap W)\}$.

Problem 2 (Sp02). Let f be a bounded, continuous, real valued function on $\mathbb{R}^2$. Define the function g on $\mathbb{R}$ by

$$g(x)=\int_{-\infty}^{\infty}\frac{f(x,t)}{1+t^2}\,\mathrm{d}t$$

Prove that g is continuous.

Problem 3 (Sp02). Prove that S_n is isomorphic to a subgroup of A_{n+2}.

Problem 4 (Sp02). Let r be a positive number and a a complex number such that $|a|\neq r$. Evaluate the integral

$$I(a,r)=\int_{|z|=r}\frac{1}{a-\bar{z}}\,\mathrm{d}z$$

where the circle $|z|=r$ has the counterclockwise orientation.

Problem 5 (Sp02). How many functions $f:\{1,2,3,4,5\}\to\{1,2,3,4,5\}$ have a image of size exactly 3 ?

Problem 6 (Sp02). Let $x_0, x_1, \ldots, x_n$ be distinct points of $\mathbb{R}$. Prove that there are unique real numbers $a_0, a_1, \ldots, a_n$ such that

$$\int_0^1 p(t)\,\mathrm{d}t=\sum_{j=0}^{n} a_j\, p\left(x_j\right)$$

for all polynomials of degree n or less.

Problem 7 (Sp02, Fa11). Let (X,ρ) and (Y,σ) be metric spaces. Assume that:

- $f, f_1, f_2, \ldots$ are bijective functions of X onto Y with inverses $g, g_1, g_2, \ldots$
- g is uniformly continuous
- $f_n \to f$ uniformly as $n\to\infty$.

Prove that $g_n \to g$ uniformly as $n \to \infty$.

Problem 8 (Sp02). Let R be a commutative ring with identity. Prove that the group R^* of units of R does not have exactly 5 elements.

Problem 9 (Sp02). Let p and q be nonconstant complex polynomials of the same degree whose zeros lie in the open disc $|z| < 1$. Prove that if $|p(z)| = |q(z)|$ for $|z| = 1$ then $q = \lambda p$ for a unimodular constant λ.

Problem 10 (Sp02). Let A and B be complex matrices of sizes 3×5 and 5×3, respectively, such that

$$AB = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Find the Jordan canonical form of BA.

Problem 11 (Sp02). Let the continuous real valued function φ on the complex plane satisfy the Sub-Mean-Value Property: for any point z_0,

$$\varphi(z_0) \leqslant \frac{1}{2\pi} \int_0^{2\pi} \varphi(z_0 + re^{i\theta})\, d\theta \qquad \text{for} \quad 0 < r < 1$$

Prove that φ obeys the Maximum Modulus Theorem: the maximum of φ over any compact set K is attained on the boundary of K.

Problem 12 (Sp02). Let G be a group of order 56 having at least 7 elements of order 7. Prove that G has only one Sylow 2-subgroup P, and that all nonidentity elements of P have order 2.

Problem 13 (Sp02). Define the function f on $\mathbb{C} \setminus [0, 1]$ by

$$f(z) = \int_0^1 \frac{\sqrt{t}}{t - z}\, dt$$

Prove that f is analytic, and find its Laurent series about ∞.

Problem 14 (Sp02). For n a positive integer, let H_n denote the n-th partial sum of the harmonic series

$$H_n = \sum_{j=1}^{n} \frac{1}{j}$$

Let $k > 1$ be an integer. Prove that

$$\log k - \frac{C}{n} < H_{nk} - H_n < \log k \qquad n = 1, 2, \ldots$$

where $\log k$ is the natural logarithm of k, and C is a constant.

Problem 15 (Sp02). 1. Determine the commutant of the $n \times n$ Jordan matrix

$$AB = \begin{pmatrix} \lambda & 1 & 0 & & 0 \\ 0 & & & & \\ & & & & 0 \\ & & & & 1 \\ 0 & & & 0 & \lambda \end{pmatrix}$$

In particular, determine the dimension of the commutant as a complex vector space.

2. What is the dimension of the commutant of the $2n \times 2n$ matrix

$$J \oplus J = AB = \begin{pmatrix} J & 0 \\ 0 & J \end{pmatrix} \; ?$$

Problem 16 (Sp02). Prove that $\displaystyle\int_0^\infty \frac{\sin x}{\sqrt{x}}\,\mathrm{d}x$ converges as an improper Riemann integral, but that $\displaystyle\int_0^\infty \frac{|\sin x|}{\sqrt{x}}\,\mathrm{d}x = \infty$.

Problem 17 (Sp02). Let p be a prime and k, n positive integers. Prove that the group $\mathrm{GL}_n(\mathbb{F}_p)$ of invertible $n \times n$ matrices over $\mathbb{F}_p$ (the field of p elements) contains an element of order p^k if and only if $n > p^{k-1}$.

Problem 18 (Sp02). Evaluate the integrals $\displaystyle I(a) = \int_{-\infty}^{\infty} \frac{\cos^3 x}{a^2 + x^2}\,\mathrm{d}x$ for $a > 0$.